Santa Clara Engineering 1851

Beyond Memorization: MELE (Memorization-resistant Evaluation for LLM Educators)

A Memorization-Resistant Evaluation of KELE-Style Socratic Teaching

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1. Introduction

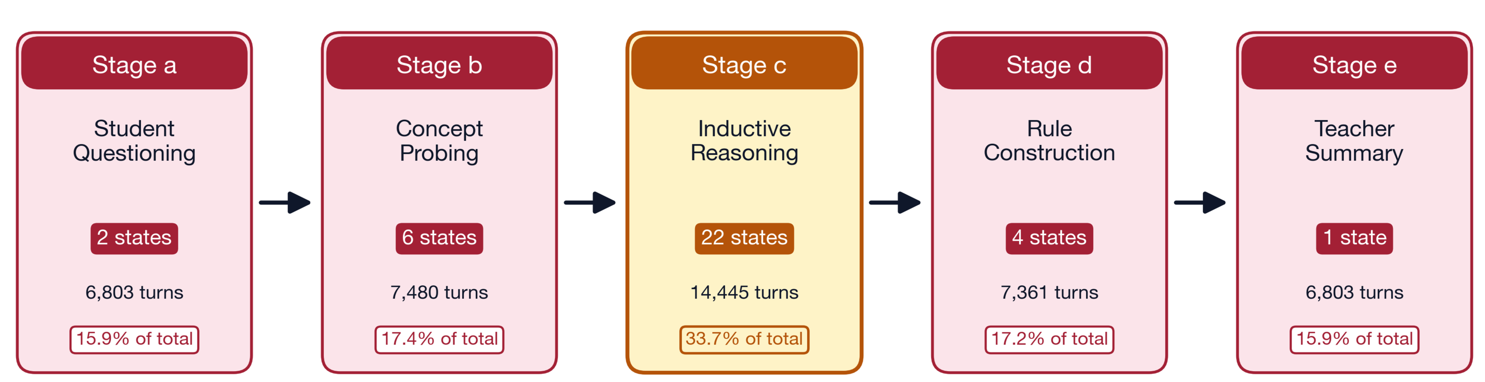
On a memorization-resistant evaluation, a 32GB GPU running an open-weight Socratic teacher overtakes the best frontier model (Anthropic's Opus 4.6) we measured and exposes that the published benchmark (reliant on ROUGE/BLEU) systematically rewards training-data memorization.

2. Problem & KELE

- **Objective**: Socratic dialogue produces measurably deeper learning than direct instruction but currently requires expert tutors at near one-to-one ratios. A scalable system would close one of education's most consequential access gaps.
- **KELE (EMNLP 2025)**: The first academic framework to structure LLMs for genuine Socratic teaching. A 5-stage / 34-state pipeline called SocRule executed by a 2-agent Consultant + Teacher architecture, separating planning (Consultant) from execution (Teacher).
- **Chosen Benchmark**: KELE's published headline rests on ROUGE / BLEU (n-gram overlap metrics adopted from machine translation) where the space of valid outputs is narrow.
- **Problem**: Socratic teaching has a vastly larger valid paraphrase space (dozens of phrasings can ask the same Socratic question correctly) so word-overlap rewards phrasing-memorization over teaching capability.
- **Outcome**: On memorization-resistant scoring, an open-weight integration on a single consumer GPU overtakes the frontier, and five converging diagnostics reveal that ROUGE/BLEU were rewarding training-data memorization the whole time.

SocRule: 5 strictly-ordered Socratic teaching stages, 34 cognitive states

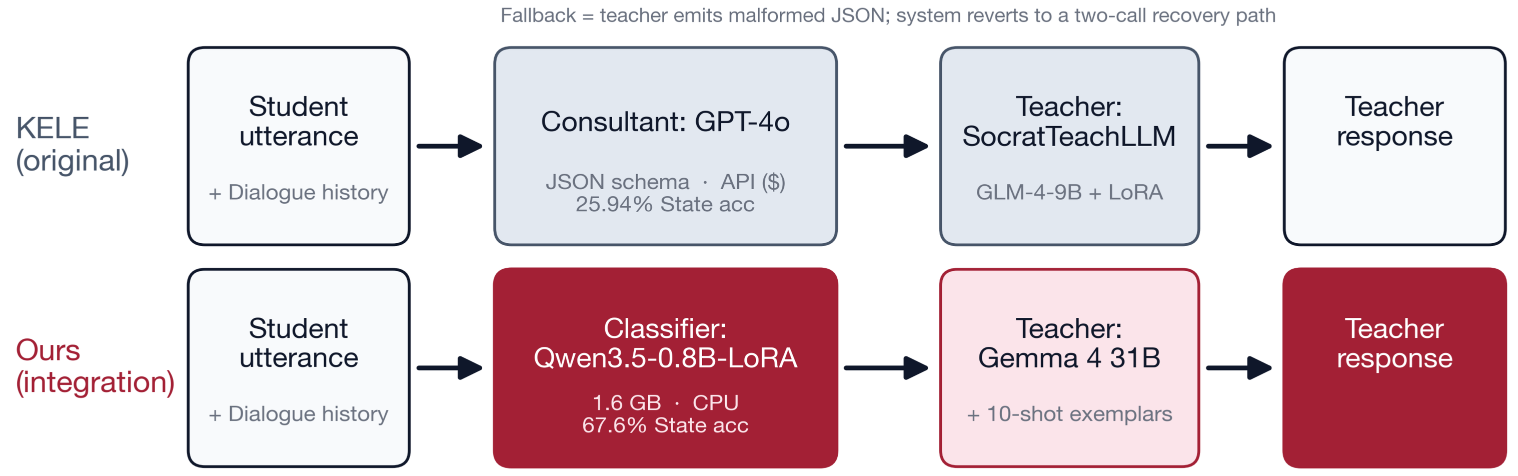
6,803 multi-turn Chinese dialogues · 42,892 annotated turns · Elementary-school science



3. Method

Architecture: LLM consultant vs. Supervised classifier

Replacing the consultant axis isolates routing from response generation



Dimension	KELE	Ours
Per-run eval API cost	~$15 / 1k turns	$0 (CPU classifier)
Consultant per-turn latency	~1.5 s (API round-trip)	~50 ms (CPU forward pass)
Teacher footprint	9B · ~19 GB GPU	31B · 32 GB GPU
Routing determinism	LLM-sampled	Deterministic

10-lever prompt-engineering tournament

500 dialogues across 10 prompt levers (n=50 each) · ranked by state accuracy · winner promoted to full n=681

#	Lever	What it does	State acc	(visual)	ROUGE-1	BLEU-4
1	Length budget	Cap tokens per turn	51.96		39.91	12.37
2	Persona	Add teacher persona prefix	51.27		39.60	12.38
3	CoT scaffold	Chain-of-thought prompt	50.89		38.98	9.55
4	Per-state exemplars	1 exemplar per 34-state	50.88		39.42	10.90
5	Negative exemplars	Anti-patterns in prompt	50.71		39.82	10.88
6	Compressed history	Summarize prior turns	47.50		38.79	9.82
7	N-best rerank	Sample 5, pick best	47.33		38.27	10.02
8	Style-matched exemplars	Match exemplars to style	46.72		42.17	12.04
9	Format retry	Retry on schema failure	46.45		39.60	10.55
10	Lexical priors	Topic glossary in prompt	43.89		39.12	9.88

4. Algorithm

Per-turn integration pipeline

Replaces KELE's LLM-consultant + JSON-schema call with a CPU classifier + deterministic SocRule lookup

Algorithm 1 · Per-turn pipeline

```
Require:  dialogue H, utterance u_t,
          classifier C, teacher T, map A, exemplars E
Ensure:   response r_t, next state s_t

1:  s_t = C(format(H, u_t))
2:  if s_t.stage < current_stage(H):
3:      s_t = snap_forward(s_t)
4:  a_t = A[s_t]
5:  E_10 = stage_balanced_sample(E, k=10)
6:  r_t = T(H, u_t, a_t, E_10)
7:  return (r_t, s_t)
```

What the algorithm guarantees

Monotonic stage progression (lines 2–3)
Socratic teaching can only move forward. snap_forward enforces this even if the classifier predicts an earlier stage.

Stage-balanced exemplar sampling (line 5)
2 exemplars per SocRule stage. Without balancing, stage c (33.7% of the corpus) would dominate the teacher's example pool.

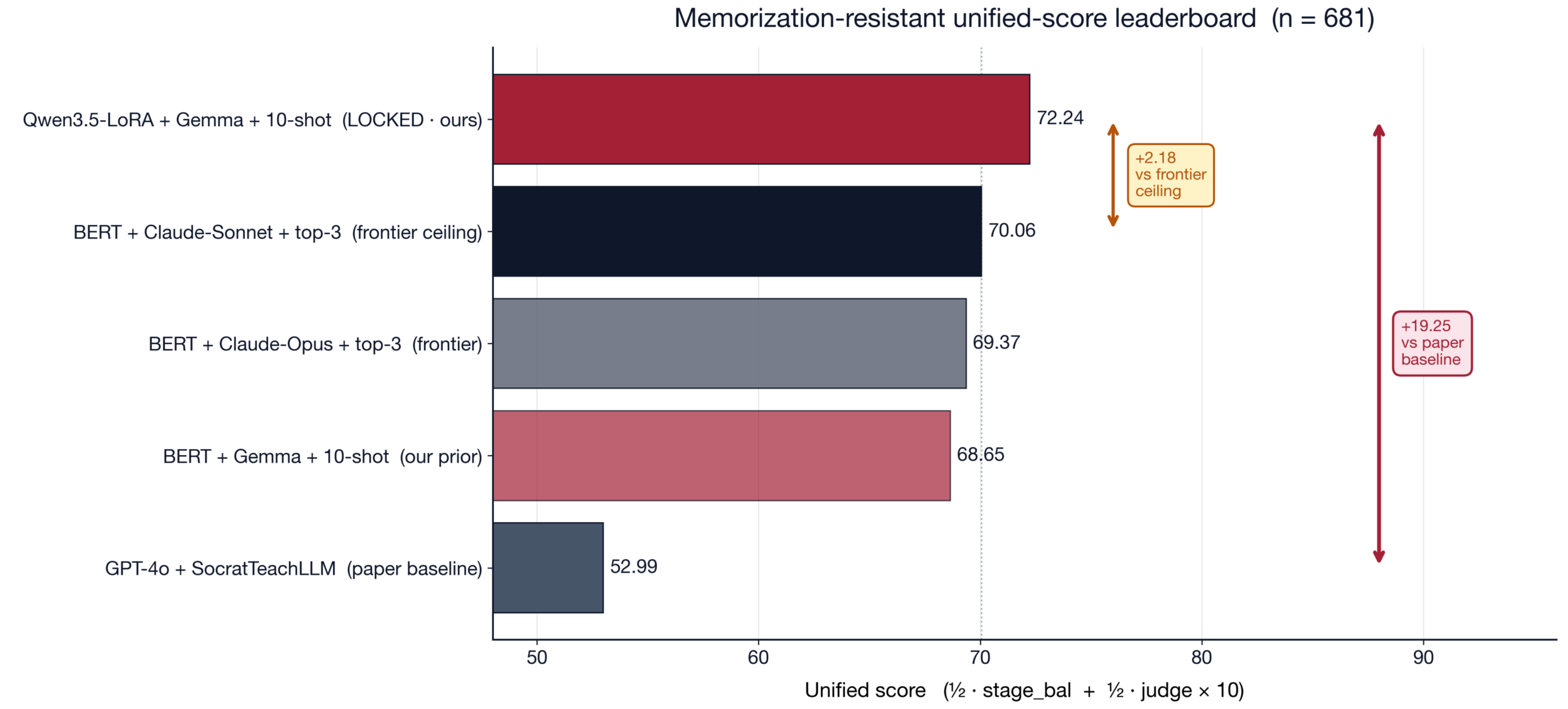
Explicit state-machine threading (line 7)
Each turn returns s_t for the next turn's monotonicity check. Dialogue is a debuggable state machine, not an LLM-implicit buffer.

End-to-end determinism (lines 1, 4, 5)
Classifier + state-action lookup + fixed-seed sampling. Same input always produces the same output. Reproducible and audit-able.

Takeaway: each numbered step encodes a structural guarantee that an LLM-consultant pipeline cannot provide

5. Evaluation Panel

Memorization-resistant unified-score leaderboard (n = 681)

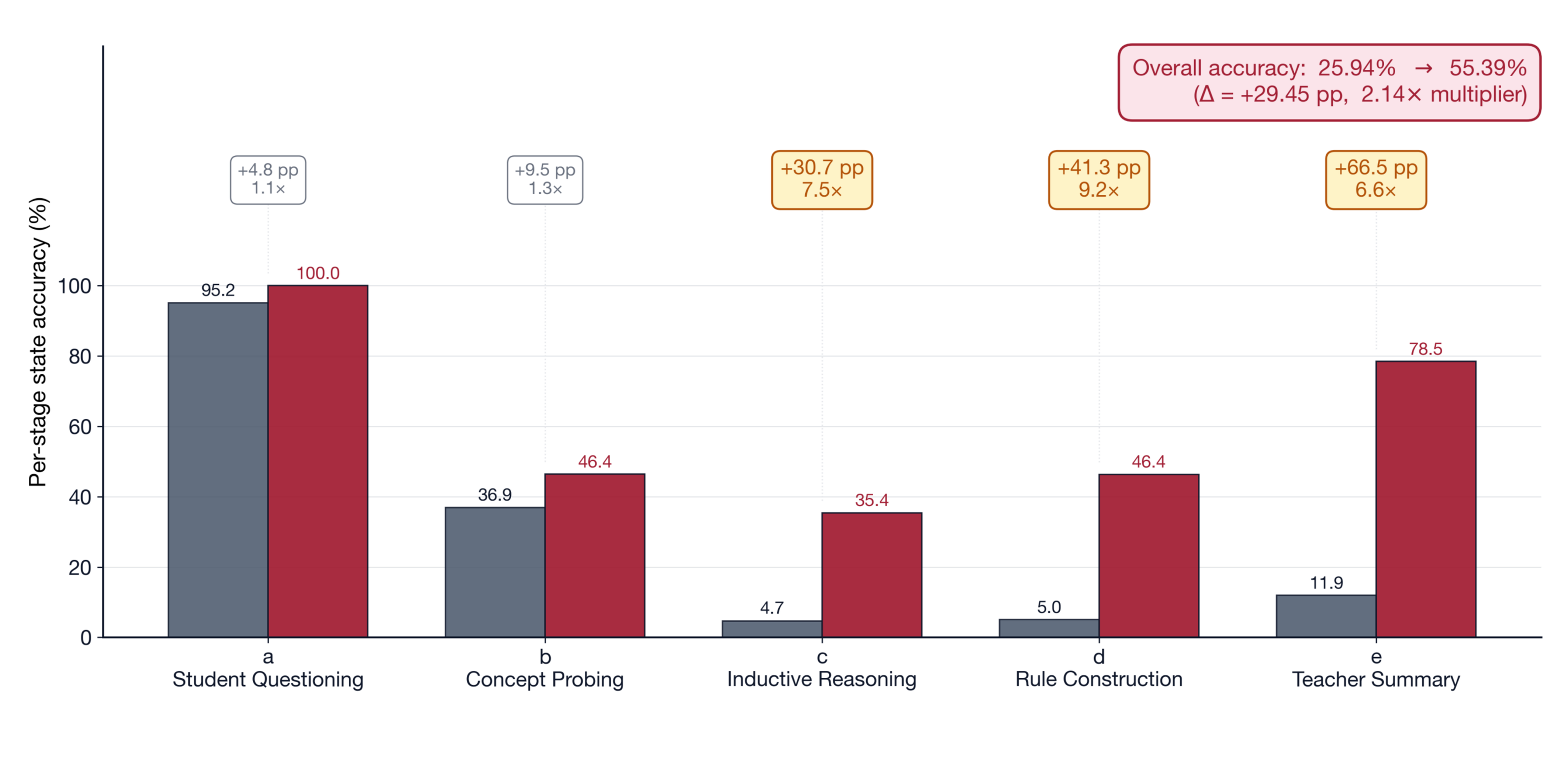


- **Headline contribution**: The unified score. It is a memorization-resistant replacement for ROUGE/BLEU as the benchmark's primary metric.
- **Why it's needed**: ROUGE/BLEU score generated text by n-gram overlap with a single reference response, which rewards memorization of phrasing rather than teaching capability.
- **How it's computed**: unified = ½ × stage balanced + ½ × (judge × 10). Both terms are paraphrase-invariant by construction.
- **Stage-balanced macro**: equal-weighted state accuracy across the 5 SocRule stages, which corrects the published metric's under-counting of the rarer closure stage.
- **LLM judge (Claude Sonnet 4.6)**: scores each teacher response 0–10 across the 4 paraphrase-invariant axes: Socratic validity, learning advancement, age-appropriateness, question-form validity.

6. Headline Result

Per-stage state accuracy: paper baseline vs. open-weight integration (n = 681)

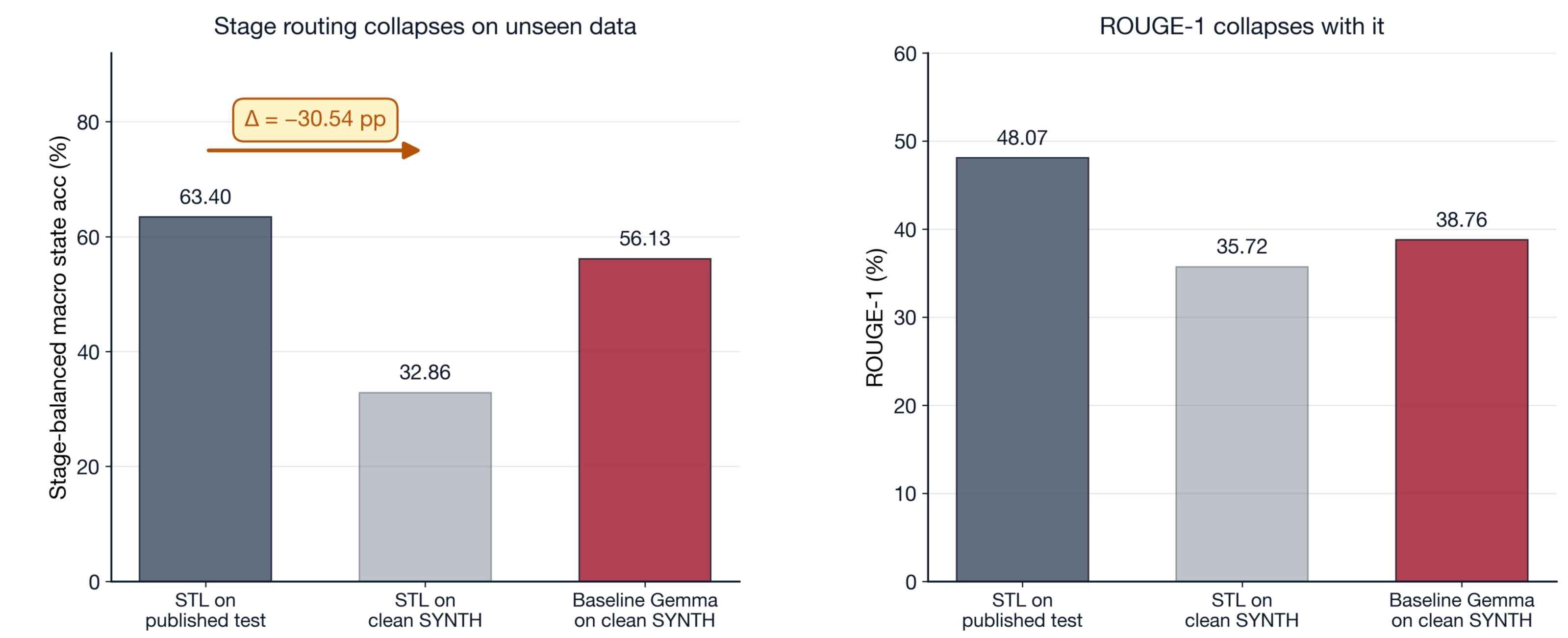
Multipliers of 6.6× – 9.2× on SocRule stages c, d, and e account for the +29.45 pp overall lift



7. Benchmark Critique

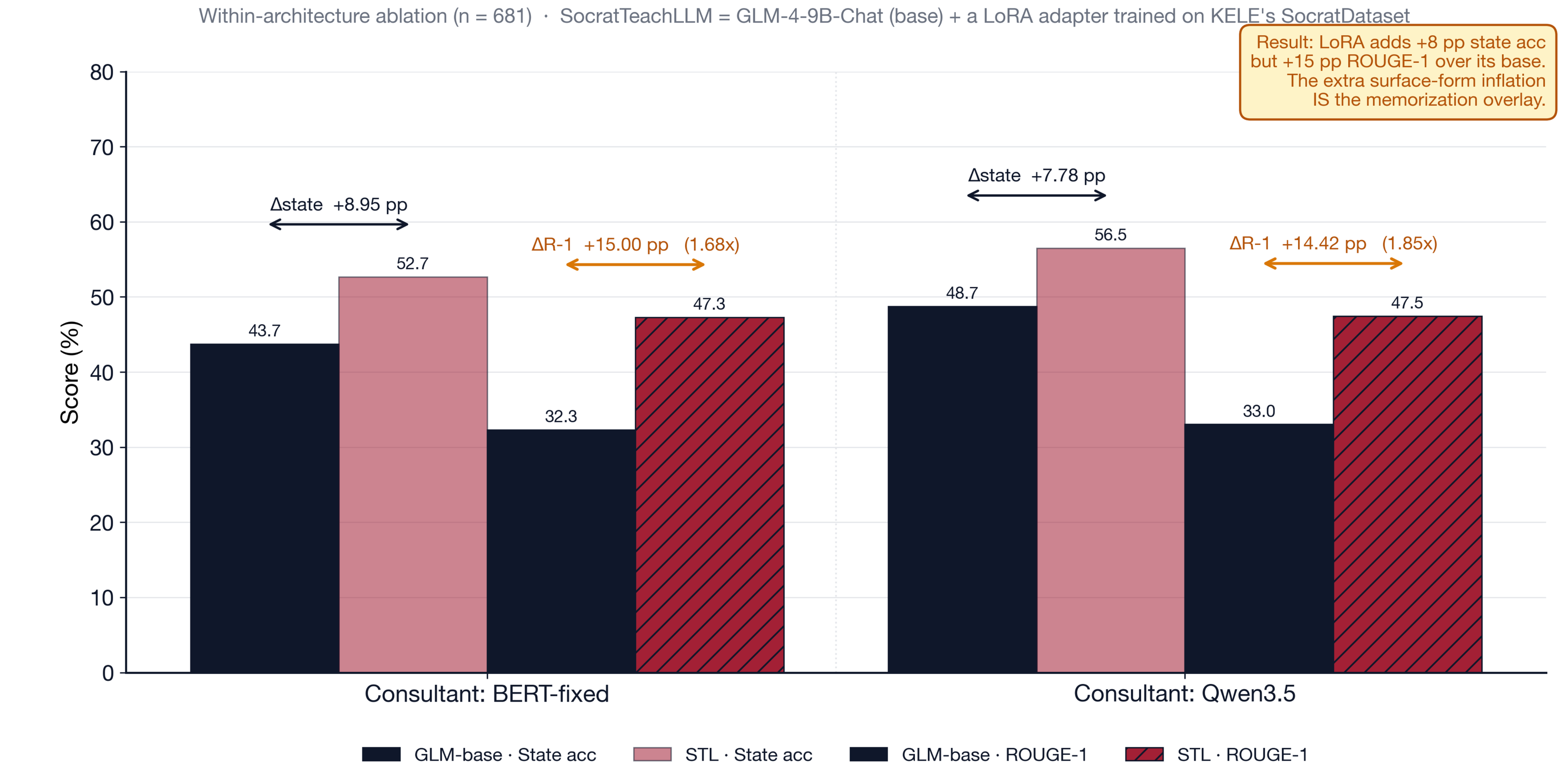
SocratTeachLLM collapses on truly unseen data

75 Sonnet-generated dialogues, disjoint from STL's training set. Even ROUGE-1 (the surface-form metric STL was trained to win) collapses on unseen data.



Did SocratTeachLLM actually learn to teach better than its base model?

Within-architecture ablation (n = 681) · SocratTeachLLM = GLM-4-9B-Chat (base) + a LoRA adapter trained on KELE's SocratDataset



8. Conclusion

1. **Architectural contribution**
A 0.8B-parameter CPU classifier replaces the LLM consultant. Routing runs 30× faster per turn (~50 ms vs. ~1.5 s API round-trip) and eliminates the 21% JSON-schema fallback rate at scale. State accuracy lifts 2.14× (+29.45 pp) and the unified score rises +19.25 points over the GPT-4o + SocratTeachLLM paper baseline. All at $0 API cost on one 32 GB consumer GPU.

2. **Methodological contribution**
Five converging diagnostics establish that ROUGE/BLEU on this benchmark reward training-data memorization, not transferable teaching capability. Three are shown above (cross-lingual translation, clean-probe collapse, within-architecture ablation); two more (rank inversion and n-gram-length scaling) are detailed in the paper.

3. **Limitations and future work**
Result is specific to this teacher, prompt, and consultant (Chinese-language SocratDataset). The +2.18 frontier-overtaking margin is supported but bounded by the n=400 variance budget. A native English-domain benchmark and bilingual classifier co-training are next.

4. **Real classrooms, today**
Elementary-school classrooms anywhere are within reach today. The full pipeline fits on a single consumer-grade GPU at near-zero per-run cost, and hardware like Nvidia's newly released N1 chip enables deployment on shared classroom rigs and individual student laptops alike. Completing the scaling work to every language and subject is how every student gets access, not only those at well-funded schools.

9. Links & References

GitHub · Hugging Face · Weights & Biases · Model Composition